

Ang Kang Heng

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Summary

AI & Automation Engineer with enterprise experience across multi-agent system design, predictive modeling, and data engineering. Proven track record of developing autonomous self-correcting agent swarms, Explainable AI models (LightGBM/SHAP), and high-throughput automation pipelines that operate without commercial API keys. Skilled at converting complex unstructured data into automated workflows that eliminate hundreds of manual hours weekly.

Experience

Zebra Technologies, CSI Specialist (Data, AI & Automation)

Penang, Malaysia

Apr 2026 – present

6 months

- Engineered a Python API using **direct SSE stream dispatch** to turn enterprise chatbots into programmatic tools without API keys, automating batch classification to reduce per-row processing from **2 minutes manually to 4.5 seconds** and saving **300+ manual hours weekly**.
- Architected the **PromptLoop multi-agent framework** using **Loop Engineering** and 5-Whys analysis, autonomously profiling raw data to refine classification prompts and compressing development cycles from **2 weeks to 3 days**.
- Engineered an **autonomous coding agent** utilizing **hash-anchored line-editing** and context compaction, with an **automated evaluator** driving self-correction loops to reduce manual code review overhead by **>80%**.
- Developed a Day-1 escalation predictive model** (LightGBM/CatBoost) achieving **>80% ROC AUC and F1 scores**, translating ML predictions into routing rules via **SHAP and Surrogate Trees** with automated drift detection.
- Isolated and resolved an agentic tool-level harness regression** in Google Gemini; built cross-model differential benchmarks disproving claims of user error to drive **Google's Product Engineering team** to deploy a platform fix.
- Architected a headless **multimodal video automation pipeline** combining **Faster-Whisper** and **Silero-VAD** for accent-resilient transcription and silence trimming, translating transcript edits into rendered video cuts via GPU acceleration.

Zebra Technologies, IT Intern (AI & Automation)

Penang, Malaysia

Oct 2025 – Apr 2026

7 months

- Engineered an 8-agent hierarchical AI prototype (LangGraph, ReAct)** to automate enterprise data synthesis; integrated **Reflective Memory (ReMe)** and an **LLM-as-Judge** validation layer to enforce self-correction and rigorously verify AI reasoning.
- Reduced global web testing cycles from several days to **90 minutes** by deploying a modular Web Assurance Suite (**Python, Playwright**) during a hybrid site migration, automating the validation of 2.9M+ legacy HTML links and ensuring 100% routing coverage for 114 modern **SPA** endpoints.
- Engineered a Human-in-the-Loop (HITL) ETL pipeline (Python, Pandas)** to bypass API constraints; utilized **LanceDB** for semantic embeddings and **JSON prompt injection** to automate data sanitization and dynamic charting.
- Increased CES categorization accuracy to **98.25%** and eliminated manual data entry by deploying **Chain-of-Thought (CoT)** prompt engineering within **Power Automate**, optimizing multilingual feedback triage.

Education

B.IT Universiti Teknikal Malaysia Melaka (UTeM), Bachelor of Information Technology (Games Technology) with Honours Oct 2022 – Apr 2026
CGPA: 3.93/4

Achievements

- **Dean's List Award (UTeM)**: Maintained a 3.93/4.00 CGPA by achieving a GPA above 3.50 for **all** semesters.
- **Gold Award & Industry Award (GDIID 2025)**: Swept three major categories for the architectural design of Astraeus Protocol, recognized by industry judges for technical excellence, algorithmic innovation, and commercial viability.
- **1st Place (Best Game) & 3rd Place (Best Poster)**, Workshop II Final Showcase: Outperformed 10+ teams in technical execution, innovation, and project communication for the development of 'Moral Nexus'.

Projects

AI Act Compliance Suite (Governance) | Python, Gemini API, SQLite Jan 2026 – Feb 2026

- **Designed** 5 modular Agent Plugins to decouple specialized AI Agents, **enabling autonomous, zero-touch compliance verification** for 3 EU legislative articles.
- **Engineered** a documentation engine parsing 20+ file types within 500k-character contexts, **automating the generation of complex visual architectures (Mermaid.js)**.
- **Standardized** system integration across frontend and backend teams by deploying a High Availability fallback framework, **guaranteeing 99.9% system uptime** and operational resilience.

Skills

Agentic AI: LangGraph, Multi-Agent Systems, LLMs & VLMs (Qwen, Phi-4), ReAct/CoT Logic, Prompt/Context Engineering, RAG, OCR, Vector Embeddings (LanceDB)

Machine Learning & Predictive Analytics: Ensemble Models (CatBoost, XGBoost, LightGBM, Random Forest), Explainable AI / XAI (SHAP, Surrogate Trees), NLP (TF-IDF, Topic Modeling)

Data Engineering & Architecture: Python (Polars, Pandas), Rust-based Parsers (Calamine), Power BI (DAX, Semantic Modeling), ETL Pipelines, Parquet, Data Sanitization

Process & Web Automation: Power Automate, Playwright (Shadow DOM, Context Pooling), Selenium, Async Python, CI/CD (GitHub Actions)

Programming Languages: Python, JavaScript, SQL, HTML/CSS

Game Engineering, 3D pipelines & Tools: Unreal Engine 5 (Blueprints, Network Replication), Blender, Git/GitHub, Firebase, Vercel, SQLite

Languages: Mandarin, English, Malay, Hokkien